

# AI5707 – Pattern Recognition and Machine Learning

## Assignment 3: Polynomial Regression

Linear Regression Experiment

Due date: August 29, 2026

### 1 Objective

In this assignment, you will learn about linear regression.

Noisy observations generated from unknown polynomial functions are shared with you. Your objective is to estimate the underlying function using polynomial regression and study the effect of polynomial degree on the fitted curve.

Write your Regression code. You can not use python packages to write code (lead to zero in code evaluation).

### 2 Dataset

The datasets are provided in the `noisy/` folder. The complete dataset can be accessed from the following Google Drive link:

**Google Drive – Dataset**

Each file corresponds to one batch:

```
noisy_0.txt  
noisy_1.txt  
noisy_2.txt  
...
```

For this assignment, use the file corresponding to your assigned batch number:

```
noisy_{batch_no}.txt
```

Each file contains two space-separated columns:

```
x    y
```

where  $x$  is the input variable and  $y$  is the corresponding noisy output.

Use only the data from your assigned file. Divide the data into Train set, Test set and Validation set. Choose the parameters of the polynomial that fit the test data set. Report the results on 'validation set.'

Record your observations separately on the experiments separately.

Note: Additional credit will be given for experiments done beyond what is stated.